Vendor exposure vs corpus coverage — reference for the firmware threat briefs

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Generated 2026-10-23 from `OTSDP/Inventory/analysis/arng_devices_raw.json` (the census refreshed 2026-10-22) and the live catalogs under `firmware-staging/*/*/*/catalog.csv`.

Census totals: **259,198 devices**, of which **5,811** fall in the `OT & Building Systems` category across **114 distinct vendors**.

A note on which number to quote: a vendor's *total* device count and its *OT* count differ, sometimes a lot. Tridium is **604 total / 568 OT**; Digital Systems Corp is **492 total / 463 OT**. The briefs quote the **OT** figure, and so does this table. Quoting the total inflates the fleet claim.

The table

"Hashes" is distinct SHA-256 count currently catalogued for that vendor in `firmware-staging`. "ST" is the number of states the devices appear in — a rough blast-radius indicator independent of raw count.

Census vendor	OT devices	States	Threat brief	Distinct hashes	Hashes / device
Texas Instruments	710	42	**none**	0	0.0
Tridium, Inc.	568	31	`tridium`	14,743	26.0
ECHELON CORP.	566	28	`echelon`	5,986	10.6
Digital Systems Corp	463	19	**none**	968	2.1
Johnson Controls, Inc.	302	6	`johnson-controls`	3,520	11.7
AutomatedLogic Corporation	253	12	`automatedlogic`	728	2.9
MERCURY SECURITY CORPORATION	252	15	`mercury-security`	1	0.0
TAC AB	252	9	`tac-vista-xenta`	267	1.1
Bosch Security Systems	252	8	`bosch-security`	2	0.0
Aiphone co.,Ltd	246	25	`aiphone`	2	0.0
VCS Video Communication Systems AG	246	15	**none**	0	0.0

Census vendor	OT devices	States	Threat brief	Distinct hashes	Hashes / device
Axis Communications AB	233	16	`axis-communications`	2,598	11.2
Digital Design Corporation	160	11	**none**	0	0.0
Honeywell GmbH	117	3	`honeywell`	0	0.0
Motorola Solutions Inc.	84	7	**none**	0	0.0
Hanwha Vision VietNam	72	4	**none**	0	0.0
Eaton Corporation	71	8	`eaton`	50,622	713.0
Genetec Inc.	66	7	**none**	0	0.0
SIEMENS BUILDING	66	1	`siemens`	572,304	**8,671.3**
OSI Security Devices	55	2	**none**	0	0.0
Apex, Inc.	39	9	**none**	0	0.0
Trane	38	4	**none**	0	0.0
AMERICAN POWER CONVERSION CORP	37	27	`apc`	20,899	564.8
SQUARE D COMPANY	35	8	`schneider`	267	7.6

What the table says that the briefs do not

1. Corpus effort is inversely correlated with fleet exposure.

Siemens holds **572,304 hashes for 66 devices** — roughly **86% of the entire corpus** for **1.1% of the OT fleet**. Eaton and APC are the same shape (713 and 565 hashes per device). Meanwhile the four vendors at 246–252 devices each — Mercury, Bosch, Aiphone and VCS — hold **1, 2, 2 and 0 hashes respectively**.

This is not an argument for deleting Siemens data; it is already collected and it is what made the file-type yield measurements possible. It *is* an argument that the next unit of effort is worth far more spent on the 246–252-device vendors than on deepening Siemens.

2. The largest OT "vendor" in the census is probably not a vendor.

Texas Instruments, 710 OT devices across 42 states — more than Tridium. TI does not sell building-automation controllers; this is almost certainly **OUI-only resolution**, where the census sees a TI network chip inside somebody else's product. Before anyone writes a TI brief, that classification needs resolving — the 710 devices are real, but they belong to some other vendor's product line, and identifying which would reassign a large block of the fleet. Same caution applies to the `Unclassified` top-category on nearly every OT vendor row.

3. Four vendors have real fleets and no brief at all.

Vendor	OT devices	States	Note
Digital Systems Corp	463	19	= **Galaxy Control Systems**. We hold **968 hashes** pulled from VirusTotal, so the material exists — only the brief is missing. Highest-value gap.
VCS Video Communication Systems AG	246	15	No firmware, no brief, not yet researched.
Digital Design Corporation	160	11	No firmware, no brief.
Motorola Solutions / Hanwha Vision / Genetec / OSI Security / Apex / Trane	84 / 72 / 66 / 55 / 39 / 38	—	Genetec is notable: it is a Mercury OEM channel, which ties to the Mercury acquisition problem.

4. `honeywell` **has a brief but zero firmware.** The staging tree holds only scaffolding. Since Honeywell resells Niagara-based controllers, the Tridium material now in hand may partially cover it — worth cross-referencing rather than acquiring separately.

5. State spread diverges from device count, and sometimes matters more.

APC appears in **27 states on 37 OT devices**; Aiphone in **25 states on 246**. Siemens Building sits in **1 state on 66**. A vendor present in 27 states is a wider lateral-movement surface than one concentrated in a single state, whatever the raw count. Tridium leads on both (568 devices, 31 states), which is why it was the right acquisition priority.

How to use this with the briefs

- Quote the **OT** column, not the total, and say "OT & Building Systems" when citing it.
- The **Hashes / device** column is a coverage-debt indicator, not a quality score. A low value on a large fleet (Mercury, Bosch, Aiphone) means the brief rests on advisories rather than on extracted firmware, and should say so.
- Counts move. The census was refreshed 2026-10-22 and vendor totals shifted; regenerate this table rather than copying these numbers forward.

Caveat on the hash counts: they measure what has been *catalogued*, which conflates acquisition with analysis depth. A vendor whose firmware is one large encrypted blob (Mercury's EP1501) scores 1 hash while still having been analysed; a vendor shipping a sprawling Windows install (Siemens TIA Portal) scores hundreds of thousands with little behavioural yield — a measured **3.0%** real-behaviour rate overall, and **0.0%** for the HTML that made up 62% of submissions. Read this column alongside the yield figures in [vt-parallel-throughput notes], not on its own.